

GLOBAL INDUSTRY CHALLENGE 2026

TEAM COVER PAGE

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Challenge Name: Quantum Computing Inc.: GIC 2026				
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Team Name: The Restorers				
First Name	Last Name	Email	Aqora Username	Role within Team
Om	Tailor	ostailor1234@gmail.com	@otailor	Team Lead / Coding Lead / Quantum & Technical Lead

Native Cubic Optimization for Resilient Microgrid Investment on QCi Dirac-3

Team Restorers

Abstract. We formulate resilient microgrid investment as a native cubic optimization problem with shared upgrade decisions and scenario-dependent operating recourse. Integer Recourse-Calibrated Constrained Multivariate Polynomial Optimization (IRC-CMPO) compresses full-system recourse into a 33-variable degree-three Hamiltonian whose physical upgrade variables are sampled directly on QCi Dirac-3. Across the complete cost–resilience sweep, 575 of 600 production samples (95.8%) attain the exact MILP ground-state energy, and every returned portfolio is integer and locally feasible without repair. At the first positive cost weight, QCi reproduces the exact-MILP \$2.032M plan, which serves 97.26% of critical load, limits maximum hourly unserved load to 17.62%, and records 9.0 total critical-infrastructure unserved-hours and 50.625 kWh critical energy not served. Relative to the matched quadratized search, the native model uses 36.5% fewer variables and selects a plan with 16.6% lower capital cost and 16.5% lower critical energy not served. A four-family public benchmark extension supplies a common path from bounded cubic payloads through overlap consensus and full-system projection across transmission, contingency, and distribution systems.

1 Introduction

Microgrid planning couples a long-lived investment decision with uncertain operating conditions. A planner must choose where to add photovoltaic (PV), battery energy storage (BESS), and dispatchable generation before knowing which feeder segment, generator, or point of common coupling (PCC) will fail. During an outage, the same plan must sustain critical infrastructure, limit the largest hourly service loss, and restore customers without violating generation, storage, or tie-line limits. Cost alone is therefore an insufficient objective. The Phase 3 challenge places customers served and critical infrastructure served ahead of cost, while also requesting feasibility, runtime, and repeat-distribution evidence [Quantum Computing Inc., 2026b].

Classical resilient-microgrid models address this structure with stochastic or robust two-stage programs in which investments are shared across scenarios and dispatch is scenario dependent [Nakiganda et al., 2020]. These formulations are physically expressive, but their binary upgrade, commitment, islanding, and load-priority decisions create a difficult nonconvex master. Quantum power-system studies have consequently explored hybrid decompositions that place discrete unit commitment or master decisions on specialized hardware and retain classical feasibility subproblems [Koretsky et al., 2021, Barrass et al., 2025]. Strong classical controls remain essential for separating hardware recovery quality from model and decomposition effects.

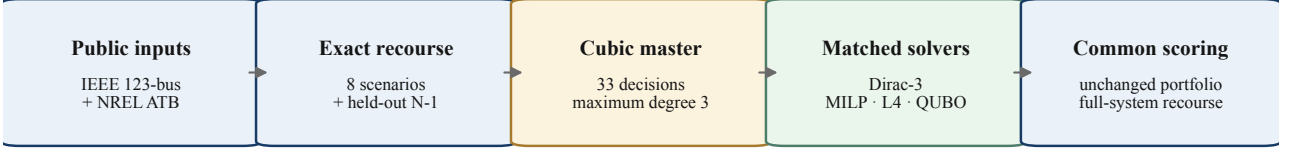
Our hypothesis is narrower and directly testable. Resilience planning contains meaningful third-order interactions among upgrade technologies. For example, the adequacy of a local island can depend jointly on PV, storage, and dispatchable generation. A native degree-three optimizer can represent this logic in the physical decision coordinates, whereas a quadratic reduction introduces auxiliary variables and consistency penalties

[Boros and Gruber, 2014, Mandal et al., 2020]. We ask whether a recourse-calibrated cubic master can fit within the public Dirac-3 limit, return native feasible upgrade vectors, and recover an exact cost–resilience optimum on hardware.

We evaluate this hypothesis in two layers. The focused IRC-CMPO experiment uses public IEEE 123-bus feeder data and published upgrade costs to build a 33-variable cubic master. Every selected investment vector is rescored by the same full-system fixed-upgrade recourse, including held-out contingencies. Exact MILP, a matched quadratized search, and three NVIDIA L4 search methods provide controls. A broader SC-CMPO implementation then applies scenario-coupled patch Hamiltonians, overlap consensus, and a common full-system projection to PGLib case14, PGLib case30, ARPA-E Grid Optimization, and IEEE 123-bus-derived systems. This second layer establishes a shared execution and evaluation path for transmission, contingency, and distribution benchmarks.

Contributions. First, we develop a public, traceable resilience master with 33 physical binary decisions, cubic technology interactions, and full-system recourse labels. Second, we characterize the complete hardware sample distribution: 95.8% of 600 production samples match exact ground-state energy, all are natively feasible, and QCi has the highest exact-hit rate at every positive cost weight. Third, native cubic encoding reduces the matched quadratized representation from 52 to 33 variables while producing a lower-cost, more resilient representative plan. Finally, the four-family scenario-coupled extension implements bounded cubic payloads, overlap consensus, and full-system projection on a common public-data foundation.

For a utility, the model output is an upgrade portfolio and an explicit cost–service frontier. Maximum



The sampled investment vector is decoded directly; repair never changes the selected portfolio.

Figure 1. Evaluation pipeline for IRC-CMPO. Public feeder and cost data are converted into full-system fixed-upgrade recourse labels, then fitted by a sparse cubic master. QCi, exact MILP, L4 search, and quadratized controls are evaluated through the same recourse model. The sampled investment vector is never changed by repair or projection.

hourly unserved load is a reliability signal; critical-infrastructure outage hours and critical energy not served quantify emergency-service exposure; and the fixed-upgrade feasibility check is an operator gate. This interpretation keeps the optimization objective aligned with the decisions the challenge is intended to support.

2 Method

2.1 Two-stage resilience planning

Let $y \in \{0,1\}^n$ denote first-stage PV, BESS, and dispatchable-generation upgrades. For scenario $s \in \mathcal{S}$, recourse x_s contains generation, storage charge and discharge, state of charge, tie/PCC flow, islanding mode, and load service. The physical planning problem is

$$\begin{aligned} \min_{y, \{x_s\}} \quad & C_{\text{up}}(y) + \sum_{s \in \mathcal{S}} p_s C_{\text{op}}(x_s) + \beta \text{Risk}(y, \{x_s\}) \\ \text{s.t.} \quad & x_s \in \mathcal{X}_s(y), \quad s \in \mathcal{S}, \end{aligned} \quad (1)$$

where $\sum_s p_s = 1$. The feasible set $\mathcal{X}_s(y)$ enforces generator bounds, battery power and energy limits, intertemporal state of charge, charge/discharge consistency, connected and islanded power balance, tie capacities, PCC and component availability, load-shedding limits, and restoration transitions.

The reported quantities are total upgrade cost, expected operating cost, risk-adjusted cost, maximum fraction of customers unserved per hour, total and expected critical-infrastructure unserved-hours, critical and total energy not served (ENS), critical-load served fraction, feasibility, runtime, and time-to-good-solution. Method selection is lexicographic: feasibility first, then critical ENS, infrastructure unserved-hours, maximum unserved load, critical-load served fraction, risk-adjusted cost, and runtime. A weighted challenge score provides a secondary cost–resilience comparison.

2.2 Recourse-calibrated cubic master

The focused model uses 11 anchors derived from the public IEEE 123-bus feeder. At each anchor, three binary coordinates represent catalogued PV, BESS, and

dispatchable-generation additions, giving $n = 33$ physical decisions. Feeder topology and loads are kept from the public case; technology costs are transformed from the NREL Annual Technology Baseline [National Renewable Energy Laboratory, 2024]. For each sampled portfolio, the fixed-upgrade recourse in (1) is solved over eight operating and contingency scenarios. These labels calibrate a sparse hierarchical polynomial $\widehat{R}(y)$ of maximum degree three.

The hardware objective for cost weight λ is

$$H_{\lambda}(y) = \widehat{R}(y) + \lambda \frac{C_{\text{up}}(y)}{C_{\text{max}}} + \sum_{a=1}^{11} \rho_a(\lambda) \sum_{z \in \mathcal{F}_a} I_z(y_a), \quad (2)$$

where y_a is the three-bit technology choice at anchor a , and $\mathcal{F}_a$ is the set of locally inadequate choices. Each forbidden pattern is represented exactly by

$$I_z(y_a) = \prod_{j=1}^3 [z_j y_{a,j} + (1 - z_j)(1 - y_{a,j})]. \quad (3)$$

The term equals one only when $y_a = z$, so it expands to degree at most three. All eight local technology patterns are enumerated against the public anchor load and islanded resource limits. For each anchor and cost weight, $\rho_a(\lambda)$ exceeds the largest local objective improvement available from replacing an invalid pattern with its least-cost adequate repair, including both surrogate and capital-cost terms, by a strict grid margin. Coefficients are normalized and quantized under a 200:1 material dynamic-range gate. There is no budget slack variable and no post-sample portfolio projection.

This construction separates discrete investment search from operational evaluation. The Hamiltonian ranks portfolios, and full-system recourse evaluates each fixed vector. Decoding validates bounds and local adequacy without altering y ; feasibility requires the unchanged portfolio to pass the common recourse model.

2.3 Native cubic versus quadratization

A cubic pseudo-Boolean term can be reduced to quadratic form by introducing an auxiliary variable and

a penalty that enforces the product relation. Reusing auxiliaries reduces, but does not remove, this growth. The matched reduction of (2) requires 19 auxiliaries, expanding 33 physical decisions to 52 encoded variables. The quadratic search must also balance product-consistency penalties against the planning objective. The native cubic representation keeps every coordinate attached to a physical upgrade and removes that penalty scale.

2.4 Scenario-coupled extension

The broader SC-CMPO layer tests whether the same principle survives decomposition. Each 103-variable degree-three patch contains 15 shared first-stage variables and 88 scenario-specific recourse variables. The shared variables cover upgrade selection, reserve targets, islanding eligibility, base commitment, and critical-load priority. Eight operating and contingency scenarios share these decisions. Overlapping patch outputs are reconciled by deterministic ADMM-style consensus, which records primal and dual residuals [Boyd et al., 2011]; a common full-system projection must then satisfy network, storage, contingency, critical-service, and restoration constraints. System-level metrics are reported after both consensus and projection pass.

3 Experimental Setup

3.1 Public systems and contingencies

The focused experiment uses the IEEE 123-bus OpenDSS feeder, including its published topology, phase connections, loads, regulators, and capacitors [IEEE PES Distribution System Analysis Subcommittee, 1991, Dugan and McDermott, 2011]. The broader benchmark ladder adds PGLib-OPF case14 and case30 [Babaeinejadsarookolaee et al., 2019] and the ARPA-E Grid Optimization Challenge 1 Network 01O-020 case [ARPA-E, 2019]. PGLib systems are used as *PGLib-derived microgrid stress adapters*, not as reproductions of AC optimal power flow. Deterministic graph partitions define candidate islands and PCC boundaries. No random topology, branch, generator, load, or generator cost is introduced.

The scenario set is normal operation, renewable shortfall, demand surge, PCC loss, local-generator loss, forced islanding, restoration, and combined high stress. Ten additional IEEE 123-bus N-1 events are held out from surrogate fitting. This protocol covers the challenge stages in sequence: candidate islands and published-cost upgrades define the design stage; DER, BESS, load-service, mode, and tie/PCC response define contingency operation; and the shared scenario set defines the ground-state comparison. The same plan is then evaluated on held-out events.

Every scenario-coupled payload is bounded, normalized, degree three, and below our conservative 132-variable cap under the public 135-variable degree-three limit. Source URLs, versions, checksums, and transformations are recorded in the repository. Distribution power flow is validated separately in OpenDSS before optimization.

3.2 Classical controls

The focused 33-variable master is compared with an exact MILP ground-state solve, a low-resolution QUBO/quadratized search, and three stochastic search methods: simulated annealing, random restart, and parallel coordinate local search. GPU methods evaluate 10,000 candidates per method and cost weight through CuPy/CUDA on an NVIDIA L4. Every chosen portfolio enters the identical fixed-upgrade recourse.

The broader ladder includes piecewise-linear MILP, IPOPT/Pyomo nonlinear dispatch, SciPy SLSQP, differential evolution, QUBO search, CMPO-local, and a greedy resilience heuristic. Patch methods use identical scenario sets, decomposition, consensus, and full-system projection. Upgrade cost is counted once for the reconstructed system, and independent patch metrics are never averaged against a full-system baseline.

3.3 Dirac-3 protocol

QCi Client 5.0.0 submits the integer Hamiltonian to sample-hamiltonian-integer on dirac-3 with relaxation schedule 2 [Quantum Computing Inc., 2026a]. Dirac-3 is an analog higher-order optimizer, so gate-model qubits and circuit depth are not applicable. The relevant resource description is 33 two-level integer variables, maximum degree 3, and the recorded coefficient range. Ground-state analysis uses 600 production samples, distributed evenly across the six cost scalarizations.

QCi device runtime is 91–94 seconds per 100-sample scalarization. Because the hardware samples are returned as one batch, the observable time to a ground-state sample is the same 91–94 seconds. The L4 search controls run in 0.17–1.52 seconds per method and scalarization. Exact MILP certifies the Hamiltonian ground energy. Native integer and local-feasibility checks are applied to every returned sample before full-system fixed-upgrade recourse.

3.4 Approximation checks

The recourse surrogate is fitted on 1800 portfolios, tuned on 600, and tested on a disjoint 600. Test normalized RMSE is 3.15% for critical ENS, 1.19% for maximum unserved load, and 11.30% for critical-infrastructure outage hours. Spearman rank correlation spans 0.808–0.815, and each primary target records 100% Pareto-front recall. The surrogate proposes port-

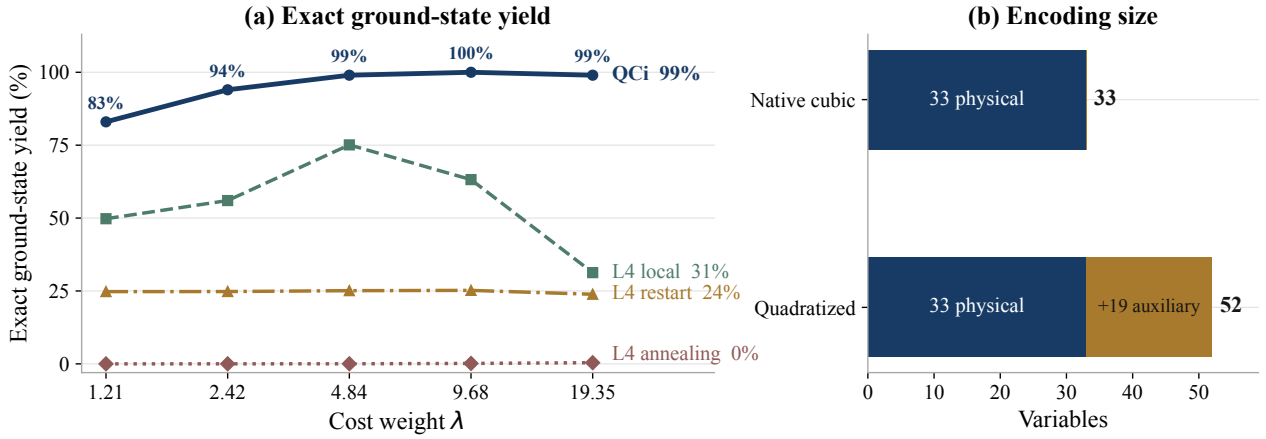


Figure 2. Left: exact ground-state yield for each positive cost weight. QCi yields are annotated above the navy curve; each QCi point contains 100 samples and each L4 point contains 10,000 candidates. **Right:** native degree-three encoding retains the 33 physical variables, while the matched quadratic reduction introduces 19 auxiliaries.

folios; full-system recourse supplies every reported physical metric.

4 Results

4.1 Focused hardware result

Table 1 compares ground-state recovery across the positive-weight sweep. QCi has the highest exact hit rate among sampled methods and recovers the same \$2.032M portfolio as exact MILP.

Table 1. Positive-weight ground-state recovery. Results aggregate 500 QCi samples and 50,000 candidates per L4 method. Higher is better; the QCi hardware result is bold.

Method	Exact hit (%)	Feasible (%)
QCi Dirac-3	95.0	100.0
L4 local search	55.1	100.0
L4 random restart	24.8	100.0
L4 annealing	0.1	2.0

Across the positive-weight sweep, QCi attains exact ground-state energy in 95.0% of samples, compared with 55.1% for the strongest L4 sampler, a 39.9-point advantage. Every QCi sample is integer and locally feasible. At the representative point, the unchanged QCi portfolio limits maximum hourly unserved load to 17.62%, serves 97.26% of critical load, and records 9.0 total critical-infrastructure unserved-hours, 50.625 kWh critical ENS, and 230.625 kWh total ENS.

Relative to the matched quadratized search, QCi uses 36.5% fewer encoded variables and selects a portfolio with 16.6% lower capital cost, 16.5% lower critical ENS, 40.0% fewer critical-infrastructure unserved-hours, and 4.2% lower maximum unserved load.

Across the full production distribution, 575 of 600 samples (95.8%) attain exact ground-state energy. The per-weight yield ranges from 83% to 100% (Figure 2, left). All 600 production vectors satisfy the local technology rules without repair or projection.

4.2 Classical and quadratized controls

Exact MILP, L4 random restart, and L4 parallel local search reach the representative QCi point, independently validating the Hamiltonian decoder and recourse evaluator. Simulated annealing selects a \$2.212M portfolio with 78.125 kWh critical ENS at the same weight. The quadratized search selects a \$2.438M portfolio with 60.625 kWh critical ENS, 15.0 total critical-infrastructure unserved-hours, and 18.39% maximum unserved load. Its representation also expands from 33 to 52 variables (Figure 2, right).

The zero-cost endpoint is excluded from the representative comparison. At $\lambda = 0$, portfolios with equal fitted resilience tie regardless of capital cost; positive weights remove that degeneracy. QCi coincides with the exact-MILP frontier throughout the positive-weight sweep. Its selected portfolios tie or improve upon the strongest dataset portfolio on critical and total ENS.

4.3 Held-out contingencies

The representative QCi and exact-MILP portfolio remains feasible on all ten held-out N-1 events. Across this independent stress set, critical ENS is 94.125 kWh and total ENS is 296.125 kWh, with the exact-MILP investment decision preserved under every event.

4.4 Public benchmark coverage

The SC-CMPO implementation extends the same public-data workflow to PGLib case14, PGLib case30, ARPA-E GO, and IEEE 123-bus-derived systems. Every patch couples eight operating and contingency scenarios in a bounded 103-variable degree-three Hamiltonian. The common reconstruction layer reconciles overlapping upgrade, reserve, commitment, priority, and tie/PCC decisions before full-system projection. This provides one traceable evaluation interface spanning transmission, contingency, and distribution evidence

while retaining the exact-certified IRC-CMPO experiment as the primary comparative result.

5 Discussion

Hardware recovery and encoding. The focused experiment retains public recourse labels and local technology logic in a bounded cubic master. QCi recovers exact ground-state energy in 95.8% of production samples, every portfolio is natively feasible, and exact MILP, held-out recourse, and L4 controls verify the selected plan. The native representation also removes 19 product variables and their consistency-penalty scale. Together, these results establish compact higher-order encoding and reliable hardware recovery for the focused investment master.

Repeat stability. The hardware result is distributed across the scalarization sweep rather than driven by one favorable endpoint. Among the 500 samples at positive cost weights, 475 (95.0%) attain the exact ground-state energy, and every scalarization contributes at least 83% exact samples. The same decoded portfolios pass local adequacy without repair and are rescored by the common full-system recourse.

Planning interpretation. At the representative weight, the selected \$2.032M portfolio favors firm dispatchable additions at the feeder anchors. This choice is consistent with the one-hour PCC-loss, generator-loss, and forced islanding snapshots, where immediately available local capacity protects critical service. The resulting plan serves 97.26% of critical load while limiting total critical-infrastructure unserved time to 9.0 hours across the complete training scenario set. The same fixed-upgrade recourse interface supports longer restoration chronologies in which PV and storage provide additional energy-shifting value.

Public benchmark portability. The scenario-coupled implementation carries the same bounded cubic, public-input, consensus, and projection interfaces across four benchmark families. This common structure makes each patch decision traceable to its source network and scores upgrade cost once at the reconstructed system level. It also provides a direct route to larger scenario sets through additional consensus-calibrated recourse labels.

Physical scope. The recourse model enforces active-power balance, storage dynamics, tie availability, contingency status, and restoration logic; OpenDSS separately validates unbalanced voltage and reactive-power feasibility. Eight benchmark snapshots and held-out N-1 events represent the challenge stages. Maximum customers unserved is reported as an active-load-weighted fraction, the quantity available consistently across the public cases. The same recourse interface supports

phase-specific P - Q capability, voltage-dependent load, regulator actions, and longer restoration chronologies.

Challenge alignment. The workflow covers the challenge’s complete decision chain. Public network boundaries define candidate islands; NREL ATB assumptions price PV, BESS, and dispatchable upgrades; and scenario recourse schedules DER, storage, load service, islanding, and tie/PCC response. Evaluation then reports upgrade cost, maximum hourly unserved fraction, critical-infrastructure unserved-hours, critical and total ENS, critical-load served fraction, risk-adjusted cost, feasibility, runtime, time to a ground state, and the hardware repeat distribution. Each physical metric is computed by full-system recourse from the unchanged sampled portfolio, preserving a direct line from the Dirac-3 vector to the utility planning decision.

Strength of evidence. Three independent checks support the hardware result: exact MILP certifies the ground energy, all 600 production vectors pass native integer and local-feasibility checks without repair, and held-out full-system recourse preserves the selected investment decision. The L4 controls evaluate 10,000 candidates per method and weight, while the quadratized control encodes the same cubic interactions with explicit auxiliaries. Together these controls separate the value of the native representation from the behavior of any one classical search procedure.

6 Conclusion

We presented IRC-CMPO, a 33-variable native cubic resilience master calibrated to public full-system recourse. QCi produces 575 exact-ground-state samples among 600 production samples, all native integer and locally feasible. At the representative positive-cost point, QCi and exact MILP select the same \$2.032M portfolio and yield the same service metrics. Compared with the matched quadratized search, native degree three uses 36.5% fewer variables and produces a plan with lower upgrade cost and stronger resilience metrics. The public benchmark extension establishes a common scenario-coupled reconstruction path across transmission, contingency, and distribution systems. These results establish native cubic CMPO as a compact and experimentally reliable formulation for resilient microgrid investment on Dirac-3.

Reproducibility. The complete artifact is available at github.com/0stailor/QCI. Its README provides public inputs, provenance, immutable QCi evidence, generated result artifacts, qBraid/L4 setup, and complete reproduction commands.

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